

RAHUL PAL

Full Stack Developer | MERN, Python and LLM/RAG Applications, Gwalior, India | +91 8962113963

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PROFESSIONAL SUMMARY

Full stack developer with 1.5+ years of experience building MERN applications for healthcare, e-commerce, investment and education products. Recent work is centered on generative AI: three GenAI products shipped, including an agentic chatbot platform with a custom RAG pipeline and reranker. On the backend, cut a client's Zoho API costs by 90% with Redis caching and handled million-record datasets in Node.js without memory crashes.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, HTML5, CSS3

Frontend: React.js (incl. React 19), Next.js, React Native, Redux Toolkit (RTK Query), TanStack Query, React Router, Tailwind CSS, Material UI, Shadcn UI, Bootstrap

Backend: Node.js, Express.js, NestJS, FastAPI, REST API design, Mongoose, Pydantic v2, JWT and cookie-based auth, OAuth, Queue-Worker Architecture, Clean Architecture

Databases: MongoDB, MySQL, Redis, Qdrant (vector database)

AI / GenAI: RAG, LangChain, LLMs, AI Agents, Tool Calling, OpenAI, DeepSeek, OpenRouter, Embeddings, Hugging Face, FastEmbed, Multi-Query Retrieval, Hybrid Search & Reranking

Cloud & Tools: AWS S3, Docker, Git, GitHub, Postman, Vercel, Netlify, Render, cPanel

Integrations: Razorpay, Stripe, Zoho API, Firebase, Expo Push Notifications

WORK EXPERIENCE

Associate Software Development Engineer | 75way Technologies Pvt. Ltd., Mohali, Punjab

Jun 2025 – Present

- Full stack work on healthcare, e-commerce and investment platforms, covering user flows from sign-up and onboarding through booking, investing and admin dashboards.
- Built booking systems, CSV bulk upload/export and sub-admin permission panels used by admin teams.
- Added a Redis cache in front of the Zoho API integration, cutting API costs by 90%, and improved MongoDB indexing to speed up responses.
- Built a custom API integration module with debounced search and export, then optimized the slow Mongoose queries and React rendering behind it.
- Fixed crashes on million-record datasets in Node.js by switching to batching, pagination and streaming instead of loading everything into memory.
- Set up AWS S3 file uploads and built animated UI components in React, Tailwind CSS and TypeScript.

Full Stack Developer | A2Developers (Remote)

Dec 2024 – Jun 2025

- Built a multi-tenant School ERP from scratch with bulk data import, automated fee management and real-time notifications, cutting manual admin work by 90–95% for the schools using it.
- Implemented bulk student onboarding through Excel import, fee assignment by class or student, and auto-generated fee receipts.
- Added bulk ID card generation with ZIP export, PDF exam report cards, and Android push notifications through Expo.
- Integrated Razorpay payments and role-based access control for employees, backed by REST APIs in Node.js/Express and MySQL.
- Built the responsive UI with React, Redux, Tailwind CSS, Material UI and Shadcn UI.

PROJECTS

Enterprise GenAI Chatbot & RAG Engine | FastAPI, React 19, Python, TypeScript, Qdrant, LangChain, DeepSeek

- Built an agentic chatbot with LangChain agents that route across OpenAI, Anthropic, Gemini, DeepSeek, OpenRouter and OpenAI, with function-calling tools (web_search, document_search), custom personas and an embeddable widget with live preview.
- Built a RAG pipeline using multi-query rewriting, semantic embeddings, and Qdrant for vector search. Added a custom reranking step that combines vector similarity with keyword matching to improve the relevance of retrieved results.
- Async FastAPI backend in a clean-architecture layout, with auto-expiring MongoDB TTL sessions, retry with exponential backoff, and a React 19 dashboard using TanStack Query.

AI Interview Assistant | Next.js, TypeScript, Firebase, Tailwind CSS, Generative AI

- Mock interview platform that generates questions based on the selected role, experience level and interview type.
- GenAI-generated feedback on technical accuracy, confidence and communication, going beyond a single score.

EDUCATION

B.Tech. in Computer Science & Engineering

IPS College of Technology and Management, Gwalior | CGPA: 8.0/10

Jul 2021 – Apr 2025